SEC Financial Risk Intelligence Skill Agent

Design Walkthrough

1. Introduction

The objective of this project is to design and implement a production-style analytics skill that evaluates financial risk using structured financial data derived from SEC filings. The system follows a deterministic analytics pipeline, ensuring that every computation, transformation, and output is auditable and reproducible. This project aligns with real-world financial analytics workflows where analysts assess company health using financial statements, ratios, trends, and anomaly detection techniques.
The system combines:

- Deterministic analytics
- Machine learning (anomaly detection)
- Rule-based scoring
- Model validation
- **LLM-powered narrative generation**

to produce **explainable, audit-ready, executive-level reports**.
Unlike traditional pipelines, this system integrates **AI-driven storytelling**, making outputs both **machine-readable and decision-ready**.
This document extends the original design () with **modern AI + reporting capabilities**.

2. Why This Domain

2.1 Relevance of Financial Analytics
Financial risk assessment is a critical function in:

- Investment decision-making
- Credit evaluation
- Corporate governance
- Regulatory compliance

Using SEC financial data provides:

- Standardized, structured data
- Real-world business context
- High analytical depth

2.2 Why SEC Data
The SEC Financial Statement Data Sets offer:

- Reliable and authoritative data
- Multi-company and time-series coverage
- Rich financial attributes (revenue, income, assets, liabilities, etc.)

This makes it ideal for:

- Trend analysis
- Ratio computation
- Risk modelling

2.3 Alignment with Assignment
This domain naturally supports:

- Deterministic workflows
- Feature engineering
- Multi-model analysis
- Structured reporting

3. System Design Overview

The Seven-Stage Pipeline
The pipeline was designed around three core principles: auditability (every number must be traceable to a formula), layered confidence (no single model has the final word), and executive accessibility (outputs must be readable without a PhD). Each stage was chosen to address a specific gap left by the previous one.

Stage 1 — Data Validation No financial analysis is trustworthy without clean inputs. Before any feature is computed, the pipeline verifies schema correctness, checks for missing or null values, detects duplicate rows, and enforces type constraints. This stage prevents garbage-in-garbage-out failures that would silently corrupt downstream risk scores

Stage 2 — Feature Engineering Raw balance-sheet line items (revenue, assets, liabilities, COGS) are transformed into financially meaningful ratios: profit margin, debt-to-equity, cash flow ratio, revenue trend, and volatility. Additionally, four Beneish M-Score sub-indices are computed — DSRI, GMI, SGI, and TATA — which serve as leading indicators of earnings manipulation.

Stage 3 — Rule-Based Risk Scoring A transparent, deterministic scoring engine evaluates six dimensions: profitability, growth, leverage, liquidity, solvency, and earnings quality. Each dimension contributes a weighted sub-score to a final 0-100 risk scale. Rule-based design was chosen here deliberately: regulators and clients can audit every scoring decision without opening a model black box

Stage 4 — Anomaly Detection (Isolation Forest) Rule-based scoring excels at known risk patterns but misses novel or compound anomalies. An Isolation Forest ML model operates on the engineered feature set to detect statistically unusual financial profiles regardless of whether they map to a predefined rule. This catches edge cases like rapid-growth companies with structurally unusual cash conversion cycles.

Stage 5 — Multi-Model Validation Neither model is assumed to be correct in isolation. A validation layer computes model agreement, anomaly strength, and a stability score to produce a final confidence-weighted decision label. When models disagree, confidence is penalised — a deliberate design choice that surfaces uncertainty rather than masking it.

Stage 6 — LLM Narrative Generation Structured scores are valuable for systems but opaque to human decision-makers. A Claude-powered narrative layer transforms the structured output into plain English: executive summary, key risk drivers, red flags, outlook, and actionable recommendations. This stage bridges the gap between quantitative rigour and human comprehension

Stage 7 — HTML Dashboard Report Design Decisions Report The final output is a production-grade HTML dashboard containing scorecards, threshold indicators, embedded charts, M-Score sections, and the LLM narrative. HTML was chosen over PDF for the primary report format because it enables interactive elements, hyperlinks to EDGAR filings, and zero-dependency rendering in any browser

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Each stage is implemented as an independent script, ensuring:

- Reusability
- Testability
- Clear separation of concerns

4. Key Design Decisions

Layer	Approach
Feature Engineering	Deterministic
Risk Scoring	Rule-based
Anomaly Detection	Machine Learning
Validation	Deterministic
Narrative	LLM-powered

4.1 Fixed vs Dynamic Dataset
Initially, the system was designed to use a fixed dataset (financial_table.csv) for consistency and reproducibility. However, this was revised to:

- Accept runtime dataset input
- Provide a default fallback dataset

This ensures compliance with assignment requirements while maintaining flexibility.
4.2 Feature Engineering Strategy
Features were chosen to represent core financial dimensions:
The pipeline consists of:

- Feature Engineering
- M-Score Feature Computation
- Anomaly Detection (Isolation Forest)
- Rule-Based Risk Scoring
- Model Validation & Decisioning
- Structured Output Generation

These features were selected because they:

- Are widely used in financial analysis
- Provide explainable insights
- Support both rule-based and ML models

4.3 Multi-Model Approach
The system uses two complementary models:
Model 1: Rule-Based M-Score Feature

- Deterministic
- Transparent
- Based on financial thresholds

Model 2: Isolation Forest (Anomaly Detection)

- Captures non-linear patterns
- Detects unusual financial behavior

4.4 Why Multi-Model?
Using a single model can be limiting:

- Rule-based models may miss hidden risks
- ML models may lack interpretability

Combining both allows:

- Better coverage
- Cross-validation of signals
- Improved confidence in results

4.5 Model Validation Strategy
To ensure reliability, results are validated using:

- Model agreement
- Stability (volatility-based)
- Anomaly strength

This prevents:

- Blind trust in one model
- Overfitting or noisy outputs

4.6 Report Design

The report was designed to be:

- Structured
- Business-friendly
- Insight-driven

Sections include:

- Executive summary
- Methodology
- Results
- Interpretation
- Recommendations

This mirrors real-world analyst reports.

4.7 Why LLM Integration?

Problem:

- Raw analytics outputs are not business-friendly

Solution:

- LLM generates:
    - Human-readable summaries
    - Decision insights
    - Executive narratives

5. What Worked Well

5.1 Structured Pipeline

Breaking the system into stages made it:

- Easy to debug
- Easy to extend
- Easy to document

5.2 Feature Engineering

The selected features provided:

- Strong signal for risk detection
- Clear business interpretation

5.3 Multi-Model Validation

Combining rule-based scoring with anomaly detection improved:

- Accuracy
- Robustness
- Explainability

5.4 Output Design

Structured outputs (JSON + HTML) ensured:

- Machine readability
- Human interpretability

6. Challenges and Fixes

6.1 Issue: Data Parsing Errors

Problem:

- SEC data in text format caused parsing issues

Fix:

- Used proper delimiter handling
- Applied robust CSV parsing

6.2 Issue: Missing or NaN Values
Problem:
- Financial datasets had missing values

Fix:
- Used median imputation instead of zero-filling
- Added validation thresholds

6.3 Issue: Imbalanced Features
Problem:
- Revenue values dominated model

Fix:
- Applied feature scaling (Standard Scaler)

6.4 Issue: Model Inconsistency
Problem:
- Rule-based and anomaly models gave conflicting outputs

Fix:
- Introduced model comparison framework
- Added validation metrics

6.5 Issue: Overly Generic Skill Design
Problem:
- Initial SKILL.md lacked specificity

Fix:
- Added exact formulas
- Defined thresholds
- Specified script interfaces

7. Limitations

- Limited to structured financial metrics
- Risk score is indicative, not predictive

8. Future Enhancements

- NLP analysis of SEC filings (MD&A)
- Market data integration
- Peer comparison
- Forecasting models
- Real-time data ingestion

9. Conclusion

This project demonstrates a complete analytics workflow applied to financial data, combining:
- Deterministic feature engineering
- Multi-model analysis
- Validation frameworks
- Professional reporting

It reflects real-world data analytics practices and highlights the importance of combining technical rigor with business interpretation.